

Enhancing Urban Mobility with V2X-Driven Smart Traffic Lights for Emergency and Public Traffic

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Abstract—Communication between vehicles and various entities, known as Vehicle-to-everything (V2X), has emerged as a key element in the advancement of intelligent transportation systems (ITS), enhancing both traffic safety and flow. In response to this, we introduce a V2X-enabled Smart Traffic Light System, called Trafficx, which is implemented using Raspberry Pi and ESP32 modules. This system allows for real-time coordination between the emergency vehicle infrastructure and ordinary civilian vehicles, helping to manage proactive traffic. Using edge computing in conjunction with wireless communication, the system offers a decentralized but efficient approach to traffic management, presenting an economical and scalable solution to the current demands of urban mobility. To evaluate the system, a realistic traffic simulation was conducted using SUMO(Simulation of Urban MObility), a renowned open-source, microscopic, and highly portable traffic simulation tool. The simulation results demonstrate a marked improvement in the response times of emergency vehicles with the proposed system. In addition to simulation, a hardware prototype was developed to identify emergency vehicles such as ambulances, fire trucks, and police cars with a custom object detection model running on the ESP32-CAM. Upon detection, the system automatically turned all other traffic signals red, granting priority to the emergency lane. The experimental results indicated a reduction in emergency vehicle wait times of approximately 60%, which ensured faster intersection clearance and improved response efficiency. The study also highlights the scalability of system and its potential integration with expansive smart city traffic management networks, especially during critical emergency scenarios.

Index Terms—V2X communication, emergency vehicle alert system, traffic light control, Raspberry Pi, ESP32, Road safety, Vehicle-to-infrastructure

I. INTRODUCTION

Today, cities are confronted with significant issues like traffic jams, extended response times to emergencies, and excessive fuel consumption, mainly because traditional traffic management systems are overly rigid. Such systems are often based on predefined fixed time intervals and do not respond to actual traffic situations, leading to inefficient signal timing and wasted delays, affecting everyone who uses the roads, including emergency responders such as ambulances, fire engines, and police cars [1], [13]. The failure of today's traffic management systems to provide emergency responders with priority in real time can lead to critical life-threatening situations and significant damage to property. To address these challenges, Vehicle-to-Everything (V2X) technology has emerged as a groundbreaking element within Intelligent Transportation Systems (ITS). V2X encompasses modes such as Vehicle-to-Vehicle (V2V), Vehicle-to-Infrastructure (V2I), and Vehicle-to-Network (V2N), allowing real-time data exchange and increased awareness at the time [1], [10]. Initial implementations using IEEE 802.11p demonstrated the viability of basic V2X communication, despite issues such as significant delays and collisions, which hinder performance in busy urban environments [1], [12]. These challenges are being overcome by next-generation cellular solutions such as LTE-V2X and 5G NR-V2X, which provide the ultra-low latency and high reliability necessary for safety applications such as Cooperative Collision Avoidance (CoCA) and priority of emergency vehicles [3], [20]. Future 6G networks are expected to offer even lower latencies and wider bandwidths,

making them ideal for managing real-time traffic signals and vehicular communication [2]. However, this added complexity necessities ensuring secure and reliable communication among diverse devices. AI-powered security systems have been proposed to detect threats, manage trust levels, and ensure the safe transmission of V2X messages [2], [9]. The integration of V2X with AI shows promise across various areas. For example, V2I communication-based predictive models can reduce intersection accidents by analyzing traffic patterns and adjusting signals in real time [4], [14]. V2X surveillance, augmented by drones, has been proposed to improve road condition assessments by combining aerial and on-road sensor inputs for a more precise identification of road damage [5]. Field trials have illustrated that the V2X system can reduce emergency vehicle traffic signal preemption and can decrease intersection wait times and improve emergency response efficiency [18]. Despite these advancements, most current systems rely on static infrastructure, expensive technology (e.g. lidar, radar), or lack complete integration with crowd-sourced traffic data [15], [16]. To address these limitations, Agarwal et al. [13] introduced a cost-effective wireless adaptive traffic control system leveraging ESP32 and Raspberry Pi technologies, utilizing travel-time data from commercial APIs like Google Maps and HERE. Their system, based on the Max-Pressure algorithm with multi intersection capabilities, achieved substantial overall delay reductions and enabled emergency vehicle preemption. This study extends these concepts, presenting a secure, AI-driven, V2X-compatible adaptive traffic control system that incorporates crowd-sourced data, V2X communications through 5G/6G, and an AI-driven trust management layer. This approach solves the triple challenges of real-time traffic management, emergency vehicle prioritization, and scalability, making it suitable for smart city applications.

II. PROBLEM STATEMENT

In recent decades, much focus has been put on minimizing emergency vehicle delays at highway intersections within the framework of intelligent transportation systems. Traditional traffic lights operate within a pre-fixed time schedule and lack the provision of responding in real-time to emergencies, leading to delays of utmost priority. This research presents the implementation of a real-time priority system for emergency vehicles with V2X capability based on low-cost hardware devices such as ESP32 and ESP32-CAM. The system detects emergency vehicles through camera feeds or wireless signals and adjusts traffic lights in real time so that they can proceed instantly. The purpose of the proposed system is to improve the efficiency of emergency response and ensure road safety by automating signal control and reducing waiting time.

III. RELATED WORK

Studies on Vehicle-to-Everything (V2X) communication and adaptive traffic systems have expanded extensively over the last decade with the aim of improving urban mobility, reducing traffic congestion, and facilitating emergency

operations. Early V2X systems, primarily using IEEE 802.11p, established the initial communication standards for Intelligent Transportation Systems (ITS) but struggled with the issues of latency, reliability, and scalability in high-traffic environments [1], [12]. Recent advances in cellular technology, such as LTE-V2X and 5G NR-V2X, have improved packet transmission and reduced latency, making them suitable for safety-critical tasks in real time such as cooperative collision avoidance (CoCA) and intersection management [3], [6], [20]. As 6G networks expand, they promise ultra low latency and extremely fast communication, yet they also introduce challenges for secure data exchange and trust management, partly addressed by AI-driven systems [2], [9]. In many studies, V2X has been integrated with smart traffic signal controls. Shahriar et al. introduced a machine learning-based V2I communication framework for predicting vehicle paths and managing intersections [4], while Evans et al. used adaptive V2X roadside units to synchronize traffic lights for autonomous electric cars [11]. Similarly, Lombard et al. applied deep reinforcement learning to manage V2X intersections during traffic surges [20]. Research on infrastructure improvements includes Varga et al.'s creation of a V2X-centric traffic signal system [15], and Czyżewski et al.'s proposal of intelligent road signs with V2X capabilities for adaptive speed management [16]. Djahel et al. highlighted V2I routing strategies for congestion mitigation [17], and Fong et al. provided a broader perspective on smart mobility infrastructure, emphasizing on cloud and vehicle connectivity [19]. Safety applications V2X to increase situational awareness: Gelbal et al. developed a flexible pedestrian avoidance system [5], and Galinski et al. demonstrated head-on collision prevention via V2V communication [6]. For emergency vehicle prioritization, Arikumar et al. engineered a highly reliable V2X-based alert system to clear paths in complex urban settings [18]. Beyond infrastructure centric solutions, Agarwal et al. proposed a cost-effective adaptive traffic signal system using crowd-sourced data, ESP32, and Raspberry Pi devices, which can reduce overall delay and facilitate emergency preemption without embedded sensors [13]. Their approach addresses a critical scalability issue, showing how real-time data from commercial APIs can enable Max-Pressure phase decisions at intersections. Collectively, these studies highlight the shift from traditional timed traffic signals to AI Enhanced, V2X-empowered, cloud-connected control systems for improved urban mobility, safety, and emergency handling, and underline the need for secure, scalable frameworks ready for widespread implementation.

IV. METHODOLOGY

A. System Overview

The proposed solution is an intelligent traffic management system based on V2X, designed to improve the speed and efficiency of emergency responses by recognizing emergency vehicles in real time and dynamically adjusting traffic signals to prioritize their movement. The system framework utilizes

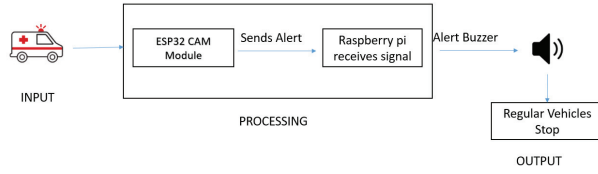


Fig. 1. Block diagram of the proposed V2X-enabled emergency vehicle detection and alert system.

both edge computing and wireless communication, offering a decentralized yet efficient traffic management mechanism. It is primarily built around an ESP32-CAM module for visual detection and a Raspberry Pi that acts as the decision-making hub. The ESP32-CAM is strategically placed at busy intersections or key traffic points to continuously capture and process images. A compact, pre-trained image classification model is incorporated into the ESP32, allowing on-device inference, thereby reducing the dependency on cloud processing and lowering latency. Upon recognizing visual indicators like distinct markings, flashing lights, or vehicle silhouettes of an emergency vehicle, the ESP32-CAM promptly sends an alert to the Raspberry Pi using a low-latency Wi-Fi connection. The Raspberry Pi reads this alert and triggers pre-programmed responses like a buzzer sound to notify drivers and pedestrians approaching the location. Alternatively, the system may be made to integrate with the current traffic light system such that the Raspberry Pi takes temporary control of normal signal patterns to provide emergency vehicles with a green tunnel. This is illustrated in Fig. 1, which shows the overall architecture of the system, including the ESP32-CAM detection unit and the Raspberry Pi-based traffic signal controller. This configuration follows Vehicle-to-Infrastructure (V2I) communication principles, which form a part of the larger Vehicle-to-Everything (V2X) system, to provide adaptive responses to traffic situations in real-time. Through decentralized decision-making through edge computing and providing low latency communication, the system is an affordable and scalable solution to contemporary urban traffic demands. Improve road safety and improve emergency response effectiveness, being a major leap in intelligent transportation systems (ITS).

B. System Architecture

The developed system is a state-of-the-art real-time prioritization system for emergency vehicles developed with the ESP32-CAM module and wireless connectivity with a Raspberry Pi. The system initially, ESP32-CAM, is trained on a dataset that includes images of emergency vehicles such as ambulances, police vehicles, and fire trucks. The training helps the module identify these vehicles with accuracy by using real-time image classification onboard. Mounted close to a traffic intersection, the ESP32-CAM continuously monitors road traffic through a live video feed. When it

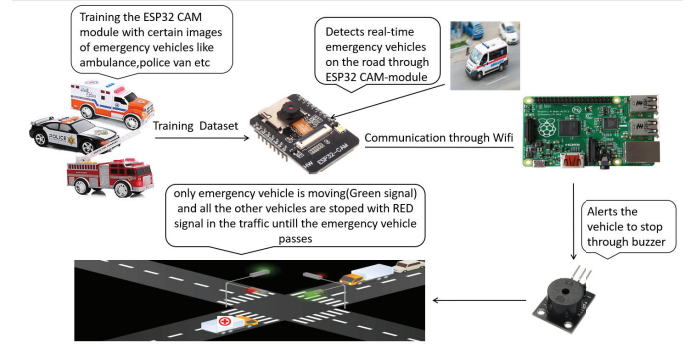


Fig. 2. System Workflow for Emergency Vehicle Detection and V2X-Controlled Traffic Management

detects an emergency vehicle, it sends a wireless signal through Wi-Fi. Upon receiving this signal, the Raspberry Pi dynamically adjusts the traffic light operation, switching the light to GREEN for the emergency vehicles path, while keeping other lanes on RED. This process is illustrated in Figure 2, which depicts the communication between the ESP32-CAM, Raspberry Pi, and traffic signal system in action. This setup allows the emergency vehicle to rapidly pass through the intersection with other vehicles behind them. To enhance safety, a buzzer is triggered to alert nearby drivers of the approaching emergency vehicle. Once the vehicle has passed, the system automatically resets the signals to normal traffic conditions. This system is lightweight, scalable, and cost-effective, offering a feasible solution for smart cities aiming to minimize emergency response times without the need for expensive infrastructure such as roadside radar or specialized vehicle sensors.

V. SIMULATION ENVIRONMENT

A. Simulation Environment Using SUMO

To test the designed V2X-based emergency vehicle priority system, a realistic traffic simulation was performed using **SUMO (Simulation of Urban MOBility)**, an open-source, microscopic and highly portable traffic simulation environment. SUMO was selected due to its robust support for custom traffic control logic, vehicle route modeling, and real-time signal control, which are essential for simulating intelligent traffic systems.

Figure 3 illustrates the central intersection configuration employed in the simulation, where all directions are initially blocked to mimic congestion, except the lane in which the emergency vehicle is located. This figure emphasizes the system's approach of selectively giving way to the emergency vehicle while restraining other vehicles to avoid conflicts.

As shown in Fig. 4, the simulated scenario is a four-way intersection, representing a typical urban traffic node with vehicles approaching from all directions. Each lane includes standard vehicle models, and traffic lights are used to manage flow based on predefined signal cycles. The simulation

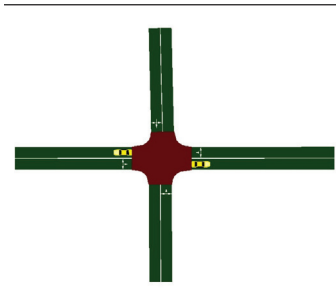


Fig. 3. "Edge-Driven Traffic Intelligence: Four-way intersection layout leveraging crowdsourced vehicle data for adaptive signal timing"

environment replicates high-density traffic to closely mimic real-world urban conditions.

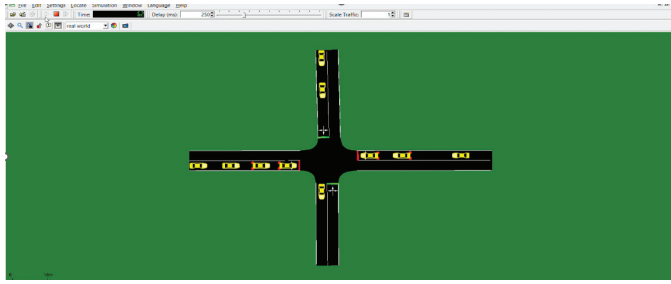


Fig. 4. SUMO-based four-way intersection layout

The core functionality demonstrated in the simulation is the detection of emergency vehicles and the dynamic control of traffic signals. An emergency vehicle (e.g., ambulance or fire engine) is introduced into one of the traffic streams. An ESP32-CAM virtual module is assumed to be placed near the intersection, tasked with detecting the approaching emergency vehicle through image processing modeled as a trigger event within the SUMO framework. The simulation results confirm that the proposed system significantly improves emergency vehicle response time. Vehicles in non-priority lanes promptly halt, and the emergency vehicle is granted uninterrupted access through the intersection. This simulation further demonstrates the system's potential for scalability and its ability to integrate with broader smart city traffic control networks, especially in time-critical emergency scenarios.

VI. RESULTS AND DISCUSSION

To evaluate the effectiveness of the proposed V2X-enabled traffic control system for emergency vehicle (EV) prioritization, a performance comparison was carried out between traditional traffic signal systems (without EV support) and our proposed system (with EV prioritization using ESP32-based wireless communication). Table I summarizes the key performance metrics.

The table illustrates that the proposed system dramatically reduces emergency vehicle waiting times from an average of 45-90 seconds to just 5-10 seconds, marking an enhancement of about 80%. The response time of the system was recorded as 2-3 seconds, enabling the rapid detection of emergency

TABLE I
PERFORMANCE COMPARISON WITH AND WITHOUT EMERGENCY
VEHICLE PRIORITY

S. No.	Parameter	Without Emergency Vehicle Priority	With Emergency Vehicle Priority (Proposed V2X System)	Improvement
1	Emergency Vehicle Waiting Time at Signal (sec)	45 – 90	5 – 10	Reduced by ~80% due to priority detection
2	Response Time of System (sec)	Not applied	2 – 3	Fast detection and alert via ESP32-CAM
3	Number of Vehicles Stopped During Alert	Not applied	3 – 6 (within 50 m radius)	Smooth clearance of path
4	Time Taken to Resume Traffic (sec)	Manual & Delayed	Automated (5 sec after Emergency Vehicle passes)	Reduced delay, improved traffic flow
5	Accuracy of Emergency Vehicle Detection (%)	Not applied	80 – 90% (based on model accuracy)	High reliability with trained dataset
6	Alert Mechanism	Manual	Buzzer & automated alert	Automated alert and control
7	Safety Improvement	Low	High	Reduces collision risk with emergency vehicle

vehicles using ESP32-CAM modules with wireless alert dissemination. Furthermore, during emergency vehicle alerts, the system affected only 3-6 vehicles within a 50-meter radius, facilitating smoother clearance. Once the emergency vehicle has passed, the system automatically resumes normal traffic flow in 5 seconds, contrasting with the delays characteristic of manual resets in traditional systems. In addition, on, the emergency vehicle detection model achieved accuracy levels between 80 and 90%, d 90%, demonstrating considerable reliability when implemented in edge devices. In general, the alert mechanism along with real-time priority control enhances safety by reducing the risk of collisions and causing minimal interference to overall traffic.

VII. "SYSTEM RESPONSE TO EMERGENCY VEHICLE DETECTION IN REAL-TIME TRAFFIC CONTROL"

The key functionality illustrated in the simulation is the identification of emergency vehicles and dynamic traffic signal control. An emergency vehicle (e.g., an ambulance or fire truck) is introduced into one of the traffic streams. A virtual ESP32-CAM module is positioned near the intersection, tasked with identifying the incoming emergency vehicle via image

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Predictions (DSP: 8 ms., Classification: 580 ms., Anomaly: 0 ms.):
Object detection bounding boxes:
Predictions (DSP: 8 ms., Classification: 580 ms., Anomaly: 0 ms.):
Object detection bounding boxes:
  fire station (0.589844) [ x: 40, y: 80, width: 8, height: 8 ]
Predictions (DSP: 8 ms., Classification: 581 ms., Anomaly: 0 ms.):
Object detection bounding boxes:
Predictions (DSP: 8 ms., Classification: 580 ms., Anomaly: 0 ms.):
Object detection bounding boxes:
Predictions (DSP: 8 ms., Classification: 580 ms., Anomaly: 0 ms.):

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Fig. 5. Real-time detection of a 'fire station' by the custom-trained model, showing bounding box, confidence score, and processing time

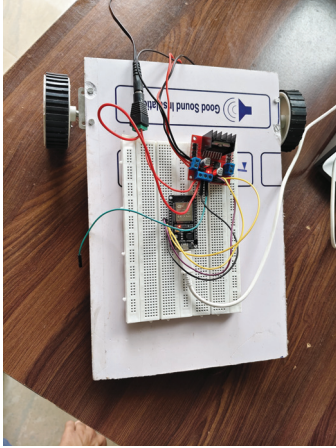


Fig. 6. Hardware setup of smart traffic control system using ESP32-CAM and Raspberry Pi

processing—modeled as a trigger event within the SUMO environment.

When the emergency vehicle is detected, simulated **V2I (Vehicle-to-Infrastructure)** communication is activated. The ESP32-CAM issues an alert (simulated as a programmatic signal) to a central controller modeled as a Raspberry Pi unit. The control logic is realized within the SUMO logic layer using **TraCI (Traffic Control Interface)**.

Upon receiving the alert, the Raspberry Pi executes the following control operations:

- **Buzzer Activation:** A virtual buzzer is activated (conceptually recorded in the simulation output).
- **Traffic Light Control:** The traffic light is immediately switched to green for the lane containing the emergency vehicle, while all opposing signals are turned red to provide a clear and safe route.

The V2X-validated smart traffic light system showcases the capability of using low-cost hardware such as ESP32 and Raspberry Pi to enhance urban traffic management. This system uses sensors or V2X communication to identify approaching vehicles, determines whether they are emergency vehicles, and then sends this information to the Raspberry Pi, which controls the traffic lights accordingly.

This strategy grants priority to emergency vehicles at intersections, reducing their wait time at signals. The arrangement not only improves emergency response times but also reduces the likelihood of accidents due to red light

violations. Wireless communication enables real-time updates without incurring significant infrastructure expenses.

Furthermore, the system's modular construction supports easy upgrades and seamless integration with existing traffic systems. Practical observations indicate that this approach can make intersections smarter and more responsive, paving the way for future integration with fully connected smart city networks.

VIII. CONCLUSION

This project introduces a cost-effective V2X-enabled intelligent traffic management system designed to prioritize emergency vehicles at intersections through real-time detection and automatic traffic light adjustments. By employing ESP32-CAM modules alongside wireless communication, the system adeptly detects emergency vehicles and adjusts the traffic signals to minimize delays and facilitate rapid passage. Simulation results indicate that this approach can cut emergency vehicle waiting times by as much as 80% while maintaining smooth traffic flow for other road users. The system's integration of automated signaling and rapid response capabilities enhances road safety and operational efficiency. It offers an affordable and scalable solution for smart cities to improve the responsiveness of emergency vehicles and shorten intersection wait times without incurring additional costs for costly infrastructure upgrades.

IX. FUTURE SCOPE

This project provides a foundation for a cost-effective and scalable V2X-capable smart traffic management system. In the future, the system can be further enhanced by adding AI-based route prediction to further optimize traffic light control in dynamic traffic conditions. The object detection model can be further developed to handle a wider variety of emergency vehicle types and enhance the detection accuracy in varying lighting and weather conditions.

Integration with city-wide traffic management centers or cloud-based systems may provide centralized monitoring and data analysis for long-term optimization. In addition, the system can be designed to support Vehicle-to-Vehicle (V2V) communication to coordinate multiple emergency vehicles that arrive at various intersections at the same time. Large-scale deployment on highways and urban areas would further prove its reliability, safety, and efficiency. The minimal hardware costs and affordability ensure that this system is ideal for smart city implementations, especially in developing markets where cheap infrastructure is key.

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